

Muhammed Shehzad Hassan

Materials Engineer | Structural Integrity | Composites & FEA

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Profile

Materials engineer (M.Sc. RWTH Aachen, Structural Integrity; expected December 2026) with a mechanical engineering background and two years of industry experience. Master's thesis at Fraunhofer IPT on a Digital Shadow for recycled carbon fibre tapes. Hands-on finite element analysis of composite and metallic structures in Ansys, with every model validated against hand calculations. Focus on damage mechanics, fracture mechanics and failure assessment.

Key Skills

Simulation	Ansys Mechanical (static structural analysis, mesh convergence), Ansys ACP (composite layups; Tsai-Wu, Puck and Hashin failure criteria), Abaqus (coursework), OpenFOAM (CFD)
Engineering methods	Classical laminate theory, stress concentration analysis, damage and fracture mechanics, analytical validation of FE results
Programming & data	Python (TODO: libraries, e.g. pandas, NumPy), C/C++
CAD	Ansys Discovery, Creo, AutoCAD, FreeCAD
Languages	English (C1), German (A1/A2, currently improving)

Experience

Master's Thesis Researcher Oct 2025 – present
Fraunhofer IPT, Aachen, Germany

- Developing a segment-based Digital Shadow framework for recycled unidirectional carbon fibre tapes, mapping material condition along the tape length
- Integrating manufacturing data streams (process parameters, laser-based geometry and quality data, camera triggers) into a traceable engineering dataset using Python
- Designed a hierarchical ID schema that traces each tape segment from recycling parameters to measured quality
- Linking recycling settings to measured tape characteristics (surface quality, dimensions) for quality-oriented evaluation

Technical Associate Jun 2020 – Jan 2021
Sutherland Global Services, Kerala, India
Resolved technical support issues; strengthened troubleshooting and customer-facing problem-solving skills.

Design Engineer Jan 2020 – Jun 2020
HomeScape Builders and Designers, Kerala, India
Supported design development and execution using CAD tools, engineering documentation and client interaction.

Mechanical Engineer Jan 2019 – Jan 2020
SJC Marine Pvt. Ltd., Kerala, India
Supported mechanical design, maintenance and documentation in a marine engineering environment.

Engineering Projects

First Ply Failure of an Open-Hole CFRP Laminate 2026
Ansys ACP, Ansys Mechanical

- Modelled a quasi-isotropic $[0/45/-45/90]_s$ carbon/epoxy laminate as layered shells; laminate stiffness (47.0 GPa) matched classical laminate theory within 0.1 %
- Predicted first ply failure at 10.1 kN (Tsai-Wu) in the 90° plies at the hole edge, confirmed by hand calculation within 10 %
- Identified and corrected a laminate-offset modelling error (88 mm spurious bending) through analytical cross-checking

Stress Concentration and Mesh Convergence Study

2026

Ansys Mechanical

- Validated a 3D FE model of a notched steel plate against analytical stress concentration theory (1 % deviation)
- Performed a mesh convergence study, identifying a mesh with 70 % fewer nodes at equal accuracy

Free-Surface Flow over a Bed Ridge (team of 3)

2023

OpenFOAM, Univ. of Oldenburg / HS Emden-Leer

- Simulated two-phase open-channel flow with a locally refined mesh (blockMesh, snappyHexMesh), run in parallel on 4 cores
- Validated against flume experiments: mean water-level deviation of about 2 cm (7 %)

Education

M.Sc. Materials Engineering (Structural Integrity)

Apr 2023 – Dec 2026 (expected)

RWTH Aachen University, Germany

Major subjects: Damage Mechanics, Fracture Mechanics, Advanced Solving Methods in Metal Forming. Master's thesis at Fraunhofer IPT.

M.Sc. Engineering Physics

Apr 2022 – Mar 2023

Carl von Ossietzky University of Oldenburg, Germany

Advanced fluid dynamics, OpenFOAM, Python for energy analysis. Transferred to RWTH Aachen to specialise in materials and structural engineering.

Master's studies in Engineering Management

Nov 2021 – Apr 2022

IUBH International University of Applied Sciences, Germany

Transferred to a public university to pursue a technical Master's degree.

Preparation for competitive examinations; relocation to Germany

Jan 2021 – Oct 2021

Preparation for Indian Armed Forces officer entrance examinations (NDA, AFCAT);
cleared the AFCAT written examination

Apr 2018 – Dec 2018

B.Tech Mechanical Engineering

2014 – 2018

TKM Institute of Technology (affiliated with Cochin University of Science and Technology), Kerala, India

Bachelor's project: pollution-sensing quadcopter with an embedded real-time sensing system.

Awards

Best Food Technology Award (2018): designed a centrifugal fat extraction concept for dairy waste treatment.